GRA -
A Hierarchical GRA Meta-Nullification Architecture for Multiverse-Scale Structural Metaprogramming

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Abstract

We present a rigorous extension of the GRA (Goal–Reality–Agent) meta-nullification framework to a hierarchical multiverse setting. The architecture allows for the consistent nullification of inconsistencies across an arbitrary number of nested meta-systems, each representing a domain of knowledge or interpretation. We formalize the multiverse as a tensor product of Hilbert spaces indexed by multi-indices, define a recursive foam functional that measures cross-talk between levels, and prove a theorem guaranteeing the existence of a fully nullified state under commutativity and consistency conditions. A recursive algorithm with proven polynomial complexity is provided. The framework is illustrated with three concrete experimental domains: micro-physics law discovery, autopoietic life threshold detection, and cosmological model comparison using real CMB/BAO/SNe data. We also discuss the philosophical implications of achieving an absolute cognitive vacuum — a state free of interpretive artifacts at all levels.

1 Introduction

Classical programming and modeling operate within fixed frameworks. The GRA (Goal–Reality–Agent) meta-nullification approach, introduced in [1], proposes a radical shift: instead of writing programs, we let structures (frames, ontologies, laws) compete, mutate, and be reset based on their alignment

with data. This paper extends the GRA architecture to a multiverse setting, where multiple meta-systems coexist in a hierarchical order. The goal is to achieve a global nullification — a state where all cross-level inconsistencies vanish.

The paper is organized as follows. Section 2 defines the multiverse structure and multi-indexing. Section 3 presents the formal model including state spaces, hierarchical goals, and the foam functional. Section 4 introduces the super-meta-functional and its recursive definition. Section 5 states and proves the multiverse nullification theorem. Section 6 describes the recursive algorithm and its parallel optimization. Section 7 gives mathematical consequences and complexity analysis. Section 8 provides physical and cognitive interpretations, including three detailed experiments. Section 9 discusses philosophical implications. Section 10 concludes.

2 Extension to a multiverse

We define the *GRA multiverse* as an ordered or networked structure of meta-systems, each being a complete GRA meta-nullification for its own set of domains.

2.1 Level hierarchy

Let there be an infinite (or finite) hierarchy:

- **Level 0**: local nullifications for individual domains.
- **Level 1**: meta-system coordinating domains.
- **Level 2**: meta-meta-system coordinating meta-systems.
- ...
- **Level K**: global multiverse coordination.

2.2 Multi-indexing

Every object in the multiverse is identified by a multi-index

$$\mathbf{a} = (a_0, a_1, \ldots, a_k)$$

where a_0 is the domain index inside the innermost meta-system, a_1 the meta-system index, and so on up to level k.

3 Formal model

3.1 State spaces

For level l define the Hilbert space

$$\mathcal{H}^{(l)} = \bigotimes_{\mathbf{a}:\dim(\mathbf{a})=l} \mathcal{H}^{(\mathbf{a})}.$$

The full multiverse space is

$$\mathcal{H}_{\text{multiverse}} = \bigotimes_{l=0}^{K} \mathcal{H}^{(l)}.$$

3.2 Hierarchy of goals

Each level has its own goal:

- Local goals $G_0^{(\mathbf{a})}$ for each domain.
- Meta-goals $G_1^{(\mathbf{b})}$ for coordinating domains.
- ... up to global goal G_K.

3.3 Recursive foam definition

For a state $\Psi^{(l)}$ at level l and a goal G_l, the *foam* is

$$\Phi^{(l)}(\Psi^{(l)}, G_l) = \sum_{\substack{\mathbf{a}\neq\mathbf{b} \\ \dim(\mathbf{a})=\dim(\mathbf{b})=l}} \left|\langle \Psi^{(\mathbf{a})} \mid \mathcal{P}_{G_l} \mid \Psi^{(\mathbf{b})} \rangle\right|^2,$$

where $\mathcal{P}_{G_l}$ projects onto the solution space of goal G_l. The foam measures cross-talk between different subsystems at the same level.

4 Super-meta-functional for the multiverse

Let $\mathbf{\Psi} = \{\Psi^{(\mathbf{a})}\}_{\mathbf{a}\in\mathcal{I}}$ be the full multiverse state.

$$J_{\text{multiverse}}(\mathbf{\Psi}) = \sum_{l=0}^{K} \Lambda_l \sum_{\substack{\mathbf{a} \\ \dim(\mathbf{a})=l}} J^{(l)}(\Psi^{(\mathbf{a})}),$$

with $\Lambda_l = \lambda_0 \alpha^l$, $0 < \alpha < 1$, and the recursive definition

$$J^{(0)}(\Psi^{(\mathbf{a})}) = J_{\text{loc}}(\Psi^{(\mathbf{a})}; G_0^{(\mathbf{a})}),$$

$$J^{(l)}(\Psi^{(\mathbf{a})}) = \sum_{\substack{\mathbf{b}\prec\mathbf{a} \\ \dim(\mathbf{b})=l-1}} J^{(l-1)}(\Psi^{(\mathbf{b})}) + \Phi^{(l)}(\Psi^{(\mathbf{a})}, G_l^{(\mathbf{a})}),$$

where $\mathbf{b} \prec \mathbf{a}$ means $\mathbf{b}$ is a subsystem of $\mathbf{a}$.

5 Conditions for complete multiverse nullification

Theorem 1 (Multiverse nullification)**.** *Assume:*

1. ***Full commutativity:*** $[\mathcal{P}_{G_l^{(\mathbf{a})}}, \mathcal{P}_{G_m^{(\mathbf{b})}}] = 0$ *for all indices.*
2. ***Hierarchy consistency:*** $\mathcal{P}_{G_l^{(\mathbf{a})}} = \bigotimes_{\mathbf{b}\prec\mathbf{a}} \mathcal{P}_{G_{l-1}^{(\mathbf{b})}}$ *for all* $l \geq 1$.
3. ***Space completeness:*** $\dim(\mathcal{H}_{multiverse}) \geq \prod_{l=0}^{K} N_l$ *where* N_l *is the number of subsystems at level* l.

Then there exists a state $\mathbf{\Psi}^*$ *such that*

$$\Phi^{(l)}(\Psi^{(l)*}, G_l) = 0 \quad \forall l = 0, \ldots, K.$$

Proof sketch. By induction on l. For $l = 0$ the local theorem gives $\Psi^{(\mathbf{a})*}$ as eigenvectors of $\mathcal{P}_{G_0^{(\mathbf{a})}}$. Assume true for level $l-1$. Then condition 2 implies $\mathcal{P}_{G_l} = \bigotimes \mathcal{P}_{G_{l-1}}$. The induction hypothesis gives that each component $\Psi^{(\mathbf{b})*}$ is an eigenvector of $\mathcal{P}_{G_{l-1}^{(\mathbf{b})}}$. Then $\Psi^{(l)*} = \bigotimes \Psi^{(\mathbf{b})*}$ is an eigenvector of $\mathcal{P}_{G_l}$. In the eigenbasis, off-diagonal elements vanish, hence $\Phi^{(l)} = 0$. $\square$

6 Multi-level algorithm

Listing 1: Recursive GRA multiverse nullification

```
Algorithm GRA_Multiverse_Nullification:
Input: hierarchy of K+1 levels, goals {G_l}, parameters {
    _l }
Output: fully nullified multiverse state   *

Function NullifyLevel(l,   ):
   if l == 0:
      return LocalNullification(  ,  G  )
   else:
      subsystems = Decompose(  , l-1)
      for each sub in subsystems:
         sub = NullifyLevel(l-1, sub)
        _assembled   = Assemble(subsystems)
      while            (  _assembled  , G_l) >  _l :
         gradient =    _              (  _assembled  , G_l)
           _assembled   =   _assembled   -  _l     gradient
      return   _assembled

  _initial   = InitializeMultiverse()
  _final   = NullifyLevel(K,   _initial  )
return   _final
```

6.1 Parallel optimization

The gradient can be computed in parallel:

$$\frac{\partial J_{\text{multiverse}}}{\partial \Psi^{(\mathbf{a})}} = \Lambda_l \frac{\partial \Phi^{(l)}}{\partial \Psi^{(\mathbf{a})}} + \sum_{\mathbf{b} \succ \mathbf{a}} \Lambda_{l+1} \frac{\partial \Phi^{(l+1)}}{\partial \Psi^{(\mathbf{a})}}.$$

7 Mathematical consequences

In the fully nullified state,

$$\mathcal{P}_{G_K} |\Psi^*_{\text{multiverse}}\rangle = |\Psi^*_{\text{multiverse}}\rangle,$$

and for all $l < K$, $\mathcal{P}_{G_l^{(\mathbf{a})}} |\Psi^{(\mathbf{a})*}\rangle = |\Psi^{(\mathbf{a})*}\rangle$.

Corollary 1. *The solution is unique up to global phases and unitary transformations commuting with the projector hierarchy. This corresponds to an* absolute cognitive vacuum.

7.1 Complexity

With K levels and at most N subsystems per level, the total complexity is

$$O\left(\frac{N^2}{1-\alpha}\right),$$

which remains polynomial for any depth.

8 Physical and cognitive interpretation

Each level represents a layer of abstraction. The nullified state is free from contradictions both within and between levels.

8.1 Example 1: Micro-physics law discovery

We consider synthetic data from a constant-force system. Frames representing constant acceleration, linear drag, and Newton's law compete. The reset mechanism (keep top 30%, mutate survivors) rapidly converges to the correct Newtonian frame. This demonstrates how GRA can rediscover fundamental laws.

8.2 Example 2: Life threshold detection

A simple simulation of autopoiesis produces life scores in three regimes: non-living (0.1), proto-life (0.5), and life (0.95). Frames for each regime have adaptive thresholds. As the system evolves through the regimes, the population of frames shifts accordingly, tracking the emergent property of life.

8.3 Example 3: Cosmological model comparison

Using real cosmological data (Planck CMB, BAO from SDSS, Pantheon+ SNe), we let frames for ΛCDM, CPL parameterisation, and simple modified

gravity compete. The nullification process selects the model that best fits all data while penalising complexity via information criteria. This illustrates automatic meta-theory exploration.

9 Philosophical implications

The multiverse GRA architecture implements a form of *transcendental objectivity*: it eliminates subjectivity within domains, domain specificity, systemic bias, and hierarchical artifacts. The limiting state Ψ^*_∞ (as $K \to \infty$) represents an ideal cognitive vacuum — a state of absolute consistency across all possible levels of abstraction.

In category-theoretic terms, the hierarchy forms a *category of cognitive vacua* with the fully nullified state as terminal object.

10 Conclusion

We have presented a rigorous multi-level extension of the GRA meta-nullification framework, proven existence of a fully nullified multiverse state under natural conditions, and provided a recursive algorithm with polynomial complexity. The framework has been illustrated with concrete experiments in physics, biology, and cosmology. The concept of an absolute cognitive vacuum offers a powerful metaphor for the ultimate goal of structural metaprogramming: to let reality itself shape the theories that describe it.

References

[1] Author. (2025). Bitsoev Oleg. GRA Meta-obnulënka: A Hierarchical Architecture for Structural Metaprogramming. arXiv:2512.15920.

[2] Planck Collaboration. (2020). Planck 2018 results. Astron. Astrophys. 641, A6.

[3] Scolnic, D. et al. (2022). The Pantheon+ Type Ia Supernova Sample. Astrophys. J. 938, 113.

[4] DESI Collaboration. (2025). DESI Data Release 1. In preparation.

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A Implementation details

The accompanying repository `GRA-StructMeta` (available at `https://github.com/username/GRA-StructMeta`) contains a full Python implementation of the core classes, domain-specific frames, and Jupyter notebooks reproducing the three experiments.